Bayesian Prior Persistence

Trust the market where it speaks, the prior where it does not

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Abstract

A desk does not re-derive its volatility surface from scratch each morning: it *persists* yesterday's surface and lets today's quotes update it. Done naively this is a dilemma — anchor hard and you *damp* a genuine overnight move; anchor softly and the unquoted wings *flap*. This note documents the Vol-Fitter's resolution: gated Bayesian persistence, where every mode adds residual rows of one universal shape and an *activation gate* turns each row off exactly where today's quotes already identify its feature. We give formal definitions of the two gating mechanisms actually shipped — the precision-ratio gate of the operator and factor modes, and the coverage-deficit weighting of the strike-gap mode — and prove the three statements the design rests on: a closed gate contributes nothing to the objective (the no-damp guarantee), the propagated precision of a signed basket is the weighted harmonic support of its legs, and persisting RR/BF as *baskets* rather than legs preserves exactly the level direction that per-leg anchoring would damp. The seven persistence modes, the diagnostics endpoint, and the transported-prior provenance are specified against the code. Two case files close the loop: the 4-vol-point overnight jump where the operator prior tracks the jumped wing while the strike anchor clings to yesterday's level, and the temporal backtest that *confirmed* the shipped hybrid default (~ 32 vol bps better held-out wing reconstruction than no prior, two-thirds of the time) while honestly convicting the pure operator modes of adding nothing at the median. The worked example, built with the production calibrator, shows an ATM jump preserved (40.7% vs the market's 39.9%) while the unquoted wing is held (40.6% vs the market-only fit's 52.6%).

Contents

1 The morning dilemma

Overnight, two things change unevenly. The at-the-money level may move sharply on fresh news, and it is *well quoted* — the liquid near-the-money options re-price immediately. The wings may have *no* fresh quotes at all, yet they still need values. A fitter faces a choice: re-fit from today's quotes only, and the model extrapolates the (now steeper) ATM skew into implausible tails — the wings *flap*; or anchor hard to yesterday's prior, and the prior pulls the *ATM* back toward yesterday — the real signal is *damped*.

Invariant

1. **Do not damp the signal.** A feature today's quotes identify is never pulled toward the prior: its gate is exactly zero, not merely small.
2. **Fill the gap.** A feature today's quotes do not identify is held at the (transported) prior, with strength that grows as coverage falls.
3. **Strictly additive.** Mode `off` is byte-identical to a plain fit; every prior block is a no-op when its target is `None`.
4. **Coupled features persist coupled.** A risk-reversal is one statement, not two leg statements; persistence must not silently decompose it.

Heuristic

The resolution is to make the prior's strength *local* and *data-aware*: strong in the wings (no quotes ⇒ trust yesterday), zero at the money (fresh quotes ⇒ trust today). This is ordinary Bayesian updating with an observation-dependent prior precision — stated as an operational rule: *do not damp the signal.*

2 The universal residual and the gate

Every persistence mode adds least-squares rows of one shape: a functional of the model minus the same functional of the transported prior, carried at a gated weight,

$$\boxed{r_j = \sqrt{\lambda_j}\,\frac{O_j(\text{model}) - O_j(\text{prior})}{\text{scale}_j},} \tag{1}$$

where O_j is a call price at a delta location, a quote operator (ATM/RR/BF/var-swap), or a local smile factor, depending on the mode (section 5).

Definition 1 (Activation gate)**.** For an observation precision $\pi^{\text{obs}} \geq 0$, a required precision $\pi^{\text{req}} > 0$ and a sharpness $\gamma > 0$,

$$\text{gap}(\pi^{\text{obs}}) = \Big(\min\big(\max\big(1 - \pi^{\text{obs}}/\pi^{\text{req}},\, 0\big),\, 1\big)\Big)^{\gamma}. \tag{2}$$

The gate is 1 where the data says nothing, decreases monotonically as the data improves, and — the decisive property — is *exactly zero* for all $\pi^{\text{obs}} \geq \pi^{\text{req}}$, not asymptotically small. γ sharpens the edge (fig. 1).

Proposition 1 (The no-damp guarantee)**.** *Fix a mode and let coordinate j satisfy $\pi_j^{obs} \geq \pi_j^{req}$, so* $\mathrm{gap}_j = 0$ *and hence $\lambda_j = 0$ in (1). Then the row $r_j \equiv 0$ for* every *model parameter vector: the objective, its gradient and its Gauss–Newton model are independent of the prior value $O_j(prior)$. In particular, with all gates closed the fit is the prior-free fit, and with mode* `off` *the fit is byte-identical to a plain fit (test-locked).*

Proof. $\lambda_j = 0$ makes r_j the zero function of θ, so it contributes 0 to the cost and a zero row to the Jacobian; nothing downstream can depend on the prior through it. The byte-identity claims are locked by `test_none_anchor_leaves_the_fit_byte_identical` and the `off`-mode routing tests. □

This is the difference between gating and shrinkage: a ridge toward the prior with a small weight still biases every coordinate a little; the gate biases identified coordinates *not at all*, and spends its entire budget on the unidentified ones.

Listing 1: The activation gate (2), verbatim from the shared vocabulary (`calib/precision.py`); matches production exactly (appendix B).

```
def activation_gap(obs, req, gamma=1.0):  # 1: data silent; 0: identified
    return np.clip(1 - obs/np.maximum(req, 1e-12), 0.0, 1.0)**gamma
```

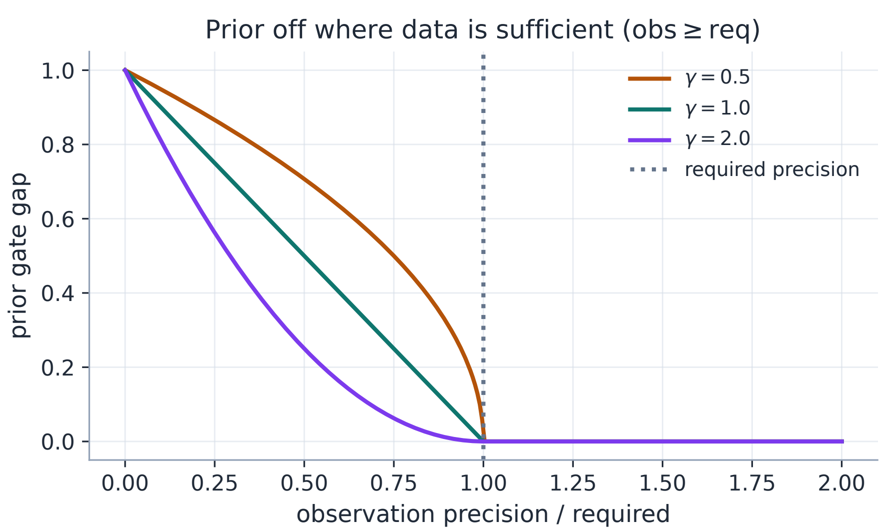

Figure 1: The activation gate (2). Past the required precision the prior weight is exactly zero — the "do not damp the signal" rule. γ controls how hard the on/off edge is.

3 Two gates, honestly distinguished

Production ships *two* gating mechanisms, and earlier revisions of this note blurred them. They answer the same question — *where should the prior pull?* — with different mathematics.

3.1 The operator/factor gate: a precision ratio

For the operator and factor modes, the observation precision of a coordinate is its *quote support*: each leg location k_a receives the Gaussian-kernel effective quote count $s_a = \sum_q \omega_q \, e^{-(k_q-k_a)^2/2b^2}$ with bandwidth $b = \texttt{priorOperatorBandwidth}$, and a basket $O = \sum_a c_a \, \sigma(k_a)$ aggregates its legs' support harmonically:

$$\pi^{\text{obs}}(O) = \Big(\sum_a \frac{c_a^2}{s_a + \varepsilon}\Big)^{-1}. \tag{3}$$

Proposition 2 (Harmonic support is the propagated basket precision)**.** *Suppose each leg vol is estimable from today's quotes with variance proportional to $1/s_a$, independently across legs. Then the implied estimate of the basket $O = \sum_a c_a\sigma(k_a)$ has variance proportional to $\sum_a c_a^2/s_a$, i.e. precision proportional to* (3)*. In particular one unquoted leg ($s_a \to 0$) drives $\pi^{obs}(O) \to 0$ however well the other leg is quoted: a risk-reversal with a missing put leg is* unidentified*, and its gate opens.*

Proof. $\mathrm{Var}\left(\sum_a c_a\hat\sigma_a\right) = \sum_a c_a^2 \,\mathrm{Var}(\hat\sigma_a) \propto \sum_a c_a^2/s_a$ by independence; take reciprocals. □

The active weights split a budget B over the open gates,

$$\lambda_j = B\,\frac{\text{gap}_j}{\sum_i \text{gap}_i}, \qquad B = \frac{\texttt{priorOperatorStrengthPct}}{100} \sum_q \omega_q, \tag{4}$$

and operators with $\text{gap}_j = 0$ are dropped from the target entirely.

Remark 1 (What the shipped gate does *not* use)**.** The required precision π^{req} is the single scalar `priorOperatorRequiredPrecision` for all operators (a per-operator schedule is declared in the design but not implemented), and the budget split (4) does *not* multiply a base prior precision. The richer vocabulary of section 3.3 — spread, freshness, transport-distance and fit-quality factors, and `active_prior_precision` — is consumed by the *graph baseline* of Note 14, not by this gate.

3.2 The strike-gap weighting: a coverage deficit

The strike-gap mode gates by a different object: a *density deficit* in log-moneyness. With ρ_{obs} the Gaussian KDE of today's quote locations (total mass = quote count, bandwidth 0.06) and ρ_{des} a desired coverage of the same mass spread over the wider anchor span, each anchor strike x_j (placed at the deltas $\{2, 5, 10, 25, 40\}\Delta$ by the Black delta map $k = \frac{1}{2}\sigma^2\tau - \sigma\sqrt{\tau}\,\Phi^{-1}(c)$) receives

$$\text{weight}_j = B_{\text{sg}} \cdot \frac{\big(\rho_{\text{des}}(x_j) - \rho_{\text{obs}}(x_j)\big)^+ \,\Delta x_j}{\sum_i \big(\rho_{\text{des}}(x_i) - \rho_{\text{obs}}(x_i)\big)^+ \,\Delta x_i}, \qquad \text{unmet} = \frac{\sum_i (\cdots)^+ \Delta x_i}{\text{desired mass}}, \tag{5}$$

with Δx_j the anchor's Voronoi cell width, and the residual is the vega-normalized *call-price* difference at x_j (the deep-wing 1/vega capped at 25× its minimum so an anchor cannot amplify noise unboundedly). Where quotes already meet the desired density the deficit is zero — the same "off where the data speaks" behaviour as definition 1, but expressed as coverage in strike space rather than a precision ratio, and never multiplied by a prior precision. The companion var-swap pull is scaled by the *unmet* fraction: a fully covered smile persists nothing, including the var-swap level.

Caution

The two mechanisms persist different *things*. Strike-gap rows anchor *absolute call prices*: if the whole vol level jumps overnight, an unquoted wing is pulled toward yesterday's *absolute* wing — correct when the level is stable, wrong across a jump. Operator/factor rows anchor *level-invariant shape* (RR, BF, curvature are unchanged by $\sigma \mapsto \sigma + c$), so the persisted wing rides *today's* level. The first case file of section 9 measures exactly this distinction — it is why the hybrid default carries the shape on operators and reserves strike anchors for the deep tail no operator covers.

3.3 The shared vocabulary and its consumers

`calib/precision.py` is the single home of the gate (2) and the scalar confidence factors

$$\begin{aligned} &\text{density} = \text{clip}(n/8,\ 0.15,\ 1), && \text{spread} = \frac{1}{1 + \text{rel spread}/0.05}, \\ &\text{freshness} = 2^{-\text{age/half-life}}\ (3\,\text{d obs},\ 30\,\text{d prior}), && \text{transport} = e^{-|h|/0.10}, \end{aligned} \tag{6}$$

with $|h| = |\log(F_{\text{now}}/F_{\text{prior}})|$ the transported forward distance. The persistence gate of section 3.1 consumes only the quote support; the *full* vocabulary — fit quality $1/\text{rms}^2$ floored at 1 vol bp, provenance tiers, and the factors (6) — prices the graph baseline of Note 14, which imports these functions from here (re-export golden-locked, so the two systems cannot drift apart).

4 Operators, factors, and why baskets

Definition 2 (Quote operators)**.** With $\sigma(k)$ a slice's implied vol, $k_{c,\delta}, k_{p,\delta}$ the call/put strikes at delta δ on the *prior* smile, and `collarSign` fixing the RR orientation:

$$\text{ATM} = \sigma(0), \qquad \text{RR}_\delta = \pm\big(\sigma(k_{c,\delta}) - \sigma(k_{p,\delta})\big), \qquad \text{BF}_\delta = \tfrac{1}{2}\big(\sigma(k_{c,\delta}) + \sigma(k_{p,\delta})\big) - \sigma(0),$$

plus VarSwap $= \sqrt{K_{\text{var}}/\tau}$. Each is a *signed basket* $O = \sum_a c_a \sigma(k_a)$ over frozen leg locations; the known set is {ATM, RR25, BF25, RR10, BF10, VarSwap}.

Definition 3 (Smile factors)**.** The factor mode replaces delta legs by ATM-local stencils of step $h =$ `priorOperatorBandwidth`: skew $= \sigma(h) - \sigma(-h)$, curvature $= \sigma(h) - 2\sigma(0) + \sigma(-h)$, leftWing $= \sigma(-3h) - \sigma(0)$, rightWing $= \sigma(3h) - \sigma(0)$ (raw differences, not divided by h) — the prior's *shape* re-expressed at the current node. Factors reuse the operator gate and target machinery verbatim.

Proposition 3 (Baskets preserve exactly the directions legs would damp)**.** *Consider one operator $O = \sum_a c_a \sigma_a$ over L legs, and compare two penalties on the leg vols $\sigma \in \mathbb{R}^L$: the basket row $r_{bsk} = \sqrt{\lambda}\,(c^\mathsf{T}\sigma - c^\mathsf{T}\sigma^{prior})$ and the per-leg rows $r_a = \sqrt{\lambda_a}\,(\sigma_a - \sigma_a^{prior})$, $a = 1..L$. The basket penalty is invariant under every leg move δ with $c^\mathsf{T}\delta = 0$ — an $(L-1)$-dimensional space that, for RR and BF, contains the* common level shift $\delta = \mathbf{1}$ *(since the coefficients sum to zero). The per-leg penalty is invariant only under $\delta = 0$: it penalizes the level shift with weight $\sum_a \lambda_a$, i.e. it damps a market-wide vol move even when the market's RR and BF are unchanged.*

Proof. r_{bsk} depends on σ only through $c^\mathsf{T}\sigma$, so its level sets are affine hyperplanes with normal c; adding any $\delta \in c^\perp$ leaves it fixed, and $\dim c^\perp = L - 1$. For RR, $c = (+1, -1)$; for BF, $c = (\tfrac{1}{2}, \tfrac{1}{2}, -1)$;

both satisfy $c^{\mathsf{T}}\mathbf{1} = 0$, so $\mathbf{1} \in c^{\perp}$. The per-leg stack has Jacobian $\mathrm{diag}(\sqrt{\lambda_a})$ of full rank, so its only invariant direction is 0, and its value at $\sigma^{\text{prior}} + t\mathbf{1}$ grows like $t^2 \sum_a \lambda_a$. $\square$

This is invariant 4 made precise, and it is why the LV route was *changed* during the build: the first design emitted synthetic per-leg quotes, which silently damped the level; the shipped design emits one signed `BasketQuote` per operator. On the local-vol surface, where the natural unknowns are call prices, the basket is transported to price space by the frozen-vega linearization $\sigma(k_a) \approx \sigma_a^{\text{prior}} + (P_a - P_a^{\text{prior}})/\mathrm{vega}_a$: weights $w_a = c_a/\mathrm{vega}_a$, target $\sum_a w_a P_a^{\text{prior}}$, tolerance $1/\sqrt{\lambda}$. It also cured a long-standing asymmetry: the SVI and Multi-Core Sigmoid (MCS) overlays, which previously received no prior at all, now consume the same operator targets as LQD.

5 The seven modes

The single source of truth is `priorPersistenceMode` (the legacy `autoLoadPrior` toggle is retired; a saved-settings blob from the old world migrates to `strike_gap` or `off`, and new installs default to `hybrid`). The resolver maps the mode to plan flags consumed by every fit path:

Mode	overlay	strike	operators	factors	tail
`off`					
`overlay`	✓				
`strike_gap`	✓	✓			
`quote_operator`	✓		✓		
`smile_factor`	✓			✓	
`hybrid` (default)	✓		✓		✓
`graph_only`	✓				

`graph_only` routes the prior exclusively into the graph baseline of Note 14 (dark nodes receive propagated innovations, no direct anchor); `hybrid`'s *tail anchor* is the strike-gap machinery restricted to the deltas strictly below the shallowest active wing operator (fallback $\{2, 5\}\Delta$), at its own budget — operators carry the shape, the tail anchor covers only what no operator reaches. Every budget is a percentage of the summed quote weights, so prior strength scales with how much data the fit actually has.

5.1 Two-pass activation

The sharpest observation precision is the *realized* fit quality — a chicken-and-egg with the fit it modifies. The opt-in `priorDataOnlyPrepass` resolves it exactly: fit once on data alone, measure each operator's realized precision, refit with priors active only on the under-observed coordinates. It is the most faithful no-damp behaviour at $\sim 2\times$ cost per node; the default single pass gates on quote support (3) with no extra fit.

6 Transport and provenance

The prior is never consumed raw. Its LQD backbone (the canonical stored object per expiry) is *transported* to the current forward under the active spot-dynamics regime (Note 12), with $h = \log(F_{\text{now}}/F_{\text{prior}})$; all operator, factor and anchor targets are evaluated on the transported smile

$w(k)$. A snapshot stores, per expiry, the LQD backbone plus the displayed-model parameters, forwards, discounts and both clocks (τ, t); persistence is to the `VolStore` database with history, and the active prior carries a provenance source (`saved` / `15min` / `close` / `none`) and a per-ticker monotone version that busts the fit cache — a fetched prior can never silently coexist with fits made under the old one.

One provenance invariant deserves its own sentence, because it is what keeps the whole persistence loop honest: *graph output is never prior input* (certification-locked). Only lit nodes with an actual calibration record enter a prior snapshot — a dark node's graph-extrapolated surface can never become a later transported-prior baseline. Without this filter the loop would echo: today's extrapolation would seed tomorrow's prior, which seeds the next extrapolation, and the graph would gradually be listening to itself. Any future prior mode that widens the snapshot set must preserve the calibrated-record filter.

7 Diagnostics

Persistence must be *inspectable*: the panel answers "is the prior pulling here, and how hard?" without reading code. Per node, `GET /smiles/{ticker}/{expiry}/prior-diagnostics` returns:

Field	Type	Meaning
`mode` / `active`	str / bool	Resolved mode; whether any prior row is live on this node.
`operators[]`	list	One row per operator (active *and* gated-off).
`operator`	str	ATM / RR25 / BF25 / VarSwap / factor name.
`priorValue`	float	The operator on the transported prior.
`obsPrecision`	float	Quote support (3).
`requiredPrecision`	float	The gate threshold in force.
`gap`	float	The gate (2): 0 = off, 1 = full pull.
`activeLambda`	float	The LSQ weight (4) actually applied.
`varSwapPriorVol` / `varSwapWeight`	float	The var-swap pull, if active.
`strikeAnchorCount`	int	Tail/strike anchors in force.

The endpoint is best-effort (it reports *inactive* rather than failing) and drives the frontend `PriorPersistencePanel`, whose table shows the same gap/λ columns. Figure 2 renders the two mechanisms of section 3 for the worked example's market: the strike-gap budget lands entirely in the unquoted wings, and the operator gate closes ATM while opening RR/BF/var-swap.

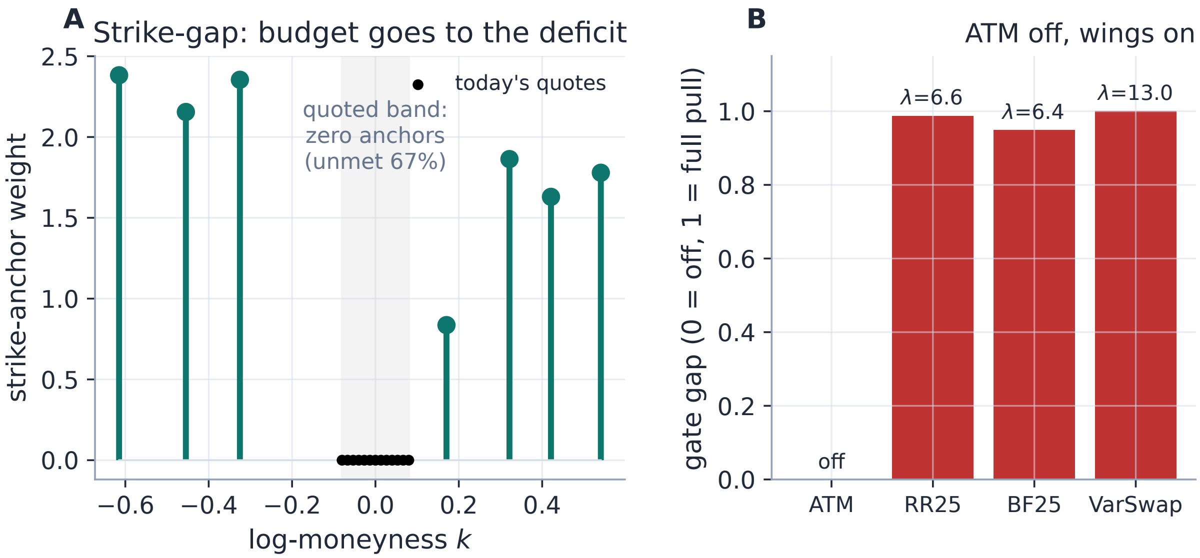

Figure 2: Activation diagnostics for a densely-ATM-quoted, wing-bare smile, computed by the production builders. Left: the strike-gap coverage deficit (5) — anchors inside the quoted band get zero budget. Right: the operator gate — ATM's quote support exceeds the requirement (gap 0, dropped from the target); the wing-dependent operators are under-observed and receive the budget.

8 Worked example: an overnight jump

Figure 3 is the dilemma made concrete, built with the production strike-gap prior and calibrator. Yesterday's prior is a calm 31.5%-ATM smile. Overnight the market jumps: today's ATM is 39.9% with a steeper skew, and *only* the ATM region ($|k| \leq 0.10$) is quoted. The market-only fit captures the ATM but extrapolates the steep skew into a wild put wing (52.6% at $k = -0.30$). The prior-persisted fit keeps the ATM at the *market* level (40.7%) because the coverage deficit (5) is zero where quotes are dense, while holding the unquoted wing at the prior (40.6% vs the prior's 41.7%) because the deficit is total where there are none. The signal is preserved; the gap is filled.

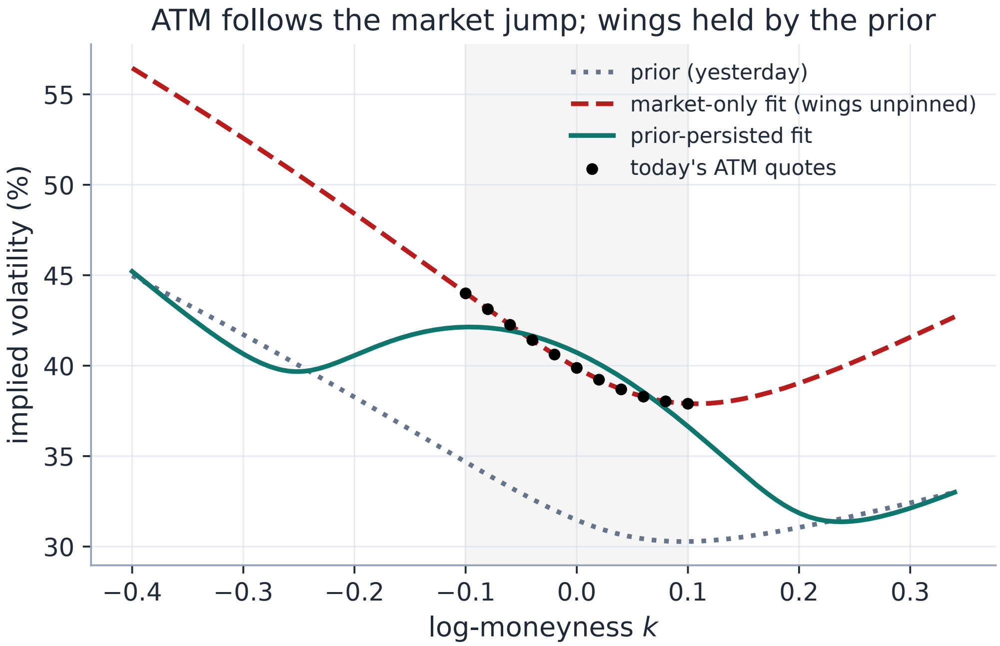

Figure 3: "Don't damp the signal." The persisted fit (solid) tracks the market ATM jump (quoted region shaded) while the unquoted wings are held by the prior (dotted), rather than flapping like the market-only fit (dashed).

Note what this example does *and does not* show: the strike-gap anchor holds the wing at yesterday's *absolute* level, which is right here because the wing genuinely had no information — but across a level jump that same absoluteness becomes the failure mode the next section measures.

9 Case files

Case file: the jump the strike anchor damped

Setup. The scenario of `test_prior_nodamp.py`: a $\tau = 0.5$ node, an overnight ATM jump of 4 vol points with the whole smile lifted, today's quotes dense at the money and absent in the wings. Operator prior (`ATM/RR25/BF25`) vs the legacy strike-gap anchor, both through the production calibrator.

Failure mode. The strike anchor pins the unquoted wing to yesterday's *absolute* wing vol — which after a level jump is the *wrong* wing: it under-shoots today's true (lifted) wing and drags the nearby smile with it.

Diagnosis. Proposition 3: per-strike price anchors penalize the common level direction $\mathbf{1}$; RR/BF baskets are invariant under it. The operator target also drops ATM automatically (`"ATM" not in op.names`) because its quote support exceeds the requirement — proposition 1 in action.

Fix. Persist *shape*, not level: the operator rows carry yesterday's RR/BF onto today's level, and the wing is reconstructed as "today's ATM + yesterday's skew/convexity".

Verdict (test-locked). The operator-prior wing lands closer to the *true jumped* wing than the strike anchor's, while the strike anchor's wing stays closer to *yesterday's* wing; the operator and factor fits match the data-only ATM to < 0.4 and < 0.6 vol points respectively — no damping. The hybrid default is precisely this division of labour: operators for shape, strike anchors only for the deep tail no operator covers.

Case file: the backtest that confirmed the default

Setup. The temporal harness: for each (asset, $T-1 \to T$) pair, fit day $T-1$'s full chain, freeze it as the active prior, thin day T to its ATM region, refit under each mode, and score the reconstructed held-out moderate wing against day T's true quotes; `off` is the baseline. Full spike regime, 1117 nodes, plus a bandwidth and var-swap probe sweep.

Verdict. The shipped **hybrid** default reconstructs the held-out wing ~ 32 vol bps better than no prior, $\sim 66\%$ of the time, and wins at *every* bandwidth and probe tested; strike-gap is a close second. The *pure* signed-operator modes do not beat `off` at the median — with the wing entirely dark, the reconstruction is driven by the tail/strike anchor, not the RR/BF rows.

Lesson. The backtest *confirmed* rather than changed the configuration: bandwidth is not a productive lever (left at 0.06), the var-swap probe stays at 1.4σ, and the hybrid's two components are both necessary — operators to avoid damping (previous case file), the tail anchor to actually rebuild a fully dark wing. Neither alone suffices; the composite does.

10 What is original here

The Bayesian framing is standard; the contributions are operational. The *activation gate* (2) turns "do not damp the signal" into one differentiable factor with a provable dead zone (proposition 1). The *harmonic basket support* (3) prices identifiability of a combination, not of legs (proposition 2). The *signed-basket* operators preserve exactly the level direction per-leg persistence would damp (proposition 3) — the theorem behind the measured case file — and incidentally cured the SVI/MCS missing-prior asymmetry. The *shared vocabulary* keeps the persistence gate and the graph baseline speaking one language without letting them silently diverge (golden re-export tests). And the whole framework is strictly additive: `off` is byte-identical to a plain fit.

11 Traceability

Table 1: Claims in this note and the code/tests that lock them.

Claim	Object	Code anchor	Test anchor
Gate: zero past requirement, monotone, γ sharpens	definition 1	`calib/precision.py`	`test_precision_shared.py::test_gap_zero_when_well_observed`, `::test_gap_monotone_and_gamma_sharpens`
No-damp / byte-identical off	proposition 1	`calib/prior.py`, `api/prior_mode.py`	`test_prior_anchor.py::test_none_anchor_leaves_the_fit_byte_identical`, `test_prior_parametric.py::test_none_arguments_are_noops`

Claim	Object	Code anchor	Test anchor
Operator math (RR sign, BF convexity)	definition 2	`calib/operators.py`	`test_operators.py::test_risk_reversal_sign_and_collar`, `::test_butterfly_positive_on_convex_smile`
Gate closes ATM on dense quotes, keeps wings	proposition 2	`calib/operators.py`	`test_operators.py::test_dense_atm_turns_off_atm_keeps_wings`, `::test_full_coverage_returns_none`
Budget splits across open gates	(4)	`calib/operators.py`	`test_operators.py::test_budget_splits_across_active_operators`
Factor stencils; hybrid tail below shallowest operator	definition 3	`calib/factors.py`	`test_prior_factors.py::test_factor_legs_stencils`, `::test_hybrid_tail_deltas_below_shallowest_operator`
Strike-gap deficit weights; vega cap	(5)	`calib/prior.py`	`test_prior_anchor.py::test_prior_targets_gating`, `::test_inv_vega_cap_bounds_tail_amplification`
Mode resolver; autoLoadPrior migration	section 5	`api/prior_mode.py`, `api/settings_persist.py`	`test_prior_mode.py::test_resolver_flags_per_mode`, `::test_migration_legacy_autoloadprior_on`
All models consume the prior (SVI/MCS asymmetry fixed)	section 4	`models/{lqd,svi_jw,sigmoid}/calibrate.py`	`test_prior_parametric.py::test_operator_prior_pulls_all_models_toward_prior_skew`
LV route emits signed baskets, not per-leg quotes	proposition 3	`api/prior_lv.py`	`test_prior_lv.py::test_rr_basket_is_a_signed_difference`, `test_prior_factors.py::test_factor_lv_targets_are_baskets`
No-damp under a level jump (case file)	section 9	`calib/operators.py`	`test_prior_nodamp.py::test_operator_prior_follows_level_and_reconstructs_jumped_wing`
Diagnostics endpoint	section 7	`api/routers/smiles.py`, `service.prior_diagnostics`	`test_prior_diagnostics.py::test_diagnostics_endpoint_ok`
Vocabulary shared with the graph (no drift)	section 3.3	`calib/precision.py`, `graph/precision.py`	`test_precision_shared.py::test_graph_reexports_shared_factors`, `::test_graph_design_point_unchanged`

Claim	Object	Code anchor	Test anchor
Persistence round-trip; transported identity	section 6	`api/priors.py`, `api/prior_transport.py`	`test_priors.py::test_priors_persist_across_restart`, `::test_transported_prior_identity_and_shift`
Graph output is never prior input (echo-free loop)	section 6	`api/priors.py`	`test_graph_dynamic_production.py` (prior-save guard)

A Hyperparameter atlas

Table 2: Prior-persistence hyperparameters.

Knob	Default	Role
Surfaced (OptionsSettings)		
`priorPersistenceMode`	`hybrid`	The mode selector (section 5); sole gate — `autoLoadPrior` is retired (migration only).
`priorAnchorWeightPct`	50	Strike-gap budget (% of summed quote weights).
`priorAnchorDeltas`	$\{2, 5, 10, 25, 40\}\Delta$	Delta locations for strike anchors.
`priorOperatorSet`	ATM, RR25, BF25, VS	Operators to persist (known: also RR10, BF10).
`priorOperatorStrengthPct`	50	Operator budget.
`priorOperatorRequiredPrecision`	1.0	Gate threshold π^{req} (one scalar for all operators).
`priorOperatorGapExponent` γ	1.0	Gate sharpness.
`priorOperatorBandwidth`	0.06	Quote-support kernel bandwidth; also the factor stencil step.
`priorOperatorCovarianceMode`	`diagonal`	Declared `diagonal`/`full`; only `diagonal` is implemented.
`collarSign`	`call_put`	RR orientation ($\pm(\sigma_c - \sigma_p)$).
`priorFactorSet`	ATM, skew, curv, VS	Factors to persist.
`priorFactorStrengthPct`	50	Factor budget.
`priorTailAnchorStrengthPct`	20	Hybrid deep-tail anchor budget.
`priorDataOnlyPrepass`	`false`	Two-pass activation (section 5.1).
Hidden (persistence path)		
`DEFAULT_BANDWIDTH`	0.06	Strike-gap KDE bandwidth.
`MAX_INV_VEGA_RATIO`	25	Cap on deep-wing 1/vega amplification.
`_VARSWAP_PROBE_STD`	1.4	Var-swap coverage probe half-width (ATM std units).
`_WING_MULT`	3	Factor wing stencils at $\pm 3h$.
`_VOL_TOL` (LV)	0.01	Vol tolerance behind LV basket/var-swap tolerances.
Hidden (shared vocabulary; consumed by the graph baseline, Note 14)		
`RMS_FLOOR`	10^{-4}	Fit-quality precision floor (1 vol bp).
`REF_ATM_QUOTES`	8	Density credit saturation.

Knob	Default	Role
`MIN_DENSITY_FACTOR`	0.15	Density credit floor.
`SPREAD_HALF`	0.05	Relative spread at which precision halves.
`OBS/PRIOR` freshness half-lives	3 / 30 d	Observation / prior age decay.
`TRANSPORT_SCALE`	0.10	Transport-distance decay scale.

Performance. The prior blocks add a handful of constant-length residual rows; every prior knob bumps the options version (one refit on change), and a fetched prior busts the fit cache. The single-pass gate is free (it reuses quote support); the two-pass prepass costs $\sim 2\times$ per node and is opt-in. Mode `off` is byte-identical to a plain fit.

B Reference implementation: the gated posterior

The gate feeds a precision-weighted Gaussian update. For one coordinate with prior mean μ_0 at gated precision λ_0 and a datum d at precision λ_d, the posterior mean is the precision-weighted average, which the least-squares stack realizes as the row $\sqrt{\lambda_0}(\theta - \mu_0)$ appended to the data rows. The listing reproduces the production `activation_gap` exactly and states the posterior it implies: $\lambda_0 \to 0$ (closed gate) gives the data, $\lambda_d \to 0$ (no data) gives the prior.

Listing 2: The gate (verbatim vs `calib/precision.py`) and the scalar posterior a gated row implements.

```
def activation_gap(obs, req, gamma=1.0):
    return np.clip(1 - obs/np.maximum(req, 1e-12), 0.0, 1.0)**gamma

def posterior(mu0, lam0, d, lam_d):    # lam0 = gated prior precision
    return (lam0*mu0 + lam_d*d) / (lam0 + lam_d)
    # lam0 -> 0: posterior -> d (no damping); lam_d -> 0: -> mu0 (gap
        filled)
```

References

[1] A. Gelman et al. *Bayesian Data Analysis.* CRC Press, 3rd ed., 2013.

[2] J. Gatheral. *The Volatility Surface.* Wiley, 2006.

[3] A. Tikhonov and V. Arsenin. *Solutions of Ill-Posed Problems.* Winston, 1977.